

CHAPTER THREE: CRYSTALLIZATION OF THE RESEARCH PROBLEM

3.0 Introduction

This chapter presents a detailed analysis of the existing fee payment system used at Koforidua Technical University and crystallizes the research problem that necessitates the development of the proposed Student Pay Mobile Application. The chapter examines the background and operational procedures of the current system, identifies its strengths and weaknesses, and compares it with related digital payment systems reviewed in Chapter Two.

The purpose of this chapter is to clearly define the specific challenges associated with the current payment process and establish the gap that the proposed system intends to address. The analysis is based on observations, existing literature, and a review of current payment technologies used in educational institutions.

3.1 Background and History of the Existing System

The management of student fee payments in higher educational institutions has evolved significantly over the years due to advancements in Information and Communication Technology (ICT). Traditionally, educational institutions relied heavily on manual payment systems where students and parents physically visited school accounts offices or designated payment centers to make payments. Under this approach, cash transactions were conducted directly with school officials, and printed receipts were issued manually as proof of payment.

Although this system provided a direct method of fee collection, it introduced numerous administrative and operational challenges. Long queues, delays in payment verification, congestion within school premises, and the risk associated with handling large amounts of cash became common issues. Furthermore, manual record-keeping increased the likelihood of errors, loss of records, and fraudulent activities.

As technology advanced, many institutions transitioned to bank-based payment systems. In this approach, students were required to visit designated banks to deposit fees into institutional accounts before presenting receipts to school authorities for verification. While this reduced congestion within school campuses, it transferred the burden to banking institutions, where long queues and delays persisted. Students also faced inconveniences associated with transportation, waiting time, and receipt management.

To improve accessibility and convenience, Koforidua Technical University later introduced a USSD-based payment system accessible through the shortcode 776123#. The system enables

students to pay tuition fees, hostel fees, transcript fees, resit fees, medical fees, graduation fees, and other institutional charges using mobile phones.

The introduction of the USSD system represented a significant improvement over previous manual and bank-based methods because it eliminated the need for physical visits to banks and enabled students to perform transactions remotely. Additionally, the system allowed students without smartphones or internet access to make payments using basic mobile phones.

Despite these improvements, the current USSD-based system still presents several limitations. The platform relies heavily on text-based interactions, lacks permanent digital transaction records, provides limited user guidance, and offers no centralized mobile interface for managing transactions. These shortcomings affect user experience, transaction reliability, and overall operational efficiency.

The increasing penetration of smartphones among university students creates an opportunity to implement a more advanced mobile-based payment platform capable of addressing the weaknesses of the current system while improving transparency, convenience, and accessibility.

3.2 Overview of the Existing System

The current payment system used by Koforidua Technical University is based on Unstructured Supplementary Service Data (USSD) technology. The system enables students to make fee payments using a mobile shortcode without requiring internet connectivity.

Students access the system by dialing the shortcode *776*102# or *776*123# on their mobile devices. After dialing the code, the system displays menu options that guide the student through the payment process. The student selects the required service, enters the student identification number, confirms payment details, and authorizes the transaction using mobile money credentials.

The system communicates with mobile money providers and banking systems to process payments. Once the transaction is completed successfully, the student receives an SMS notification confirming the payment.

Although the system simplifies remote payments, it operates mainly as a transactional platform rather than a comprehensive financial management solution. It lacks advanced functionalities such as downloadable receipts, transaction history management, user dashboards, and real-time account monitoring.

3.3 How the Existing System Operates

The operation of the current USSD payment system follows several sequential processes that facilitate communication between the student, mobile money service provider, bank, and university payment platform.

3.3.1 Payment Process

The payment process involves the following steps:

1. The student dials the USSD shortcode *776*102# or *776*123#.
2. The system displays payment menu options.
3. The student selects the preferred payment category.
4. The student enters the student identification number.
5. The system retrieves and displays student details for confirmation.
6. The student enters the payment amount.
7. The system forwards the request to the mobile money service provider.
8. The student authorizes the transaction using a mobile money PIN.
9. The mobile money platform processes the transaction through the bank.
10. Payment details are stored in the database.
11. The student receives an SMS notification confirming the transaction.

3.3.2 Components of the Existing System

The current USSD payment system consists of the following major components:

1. Student Interface

This is the user interaction layer where students dial the USSD shortcode and navigate through payment menus.

2. USSD Gateway

The USSD gateway handles communication between the mobile network operator and the payment system. It processes user requests and manages session-based interactions.

3. Payment Processing Module

This component handles payment authorization and communication with mobile money providers and banking systems.

4. Database System

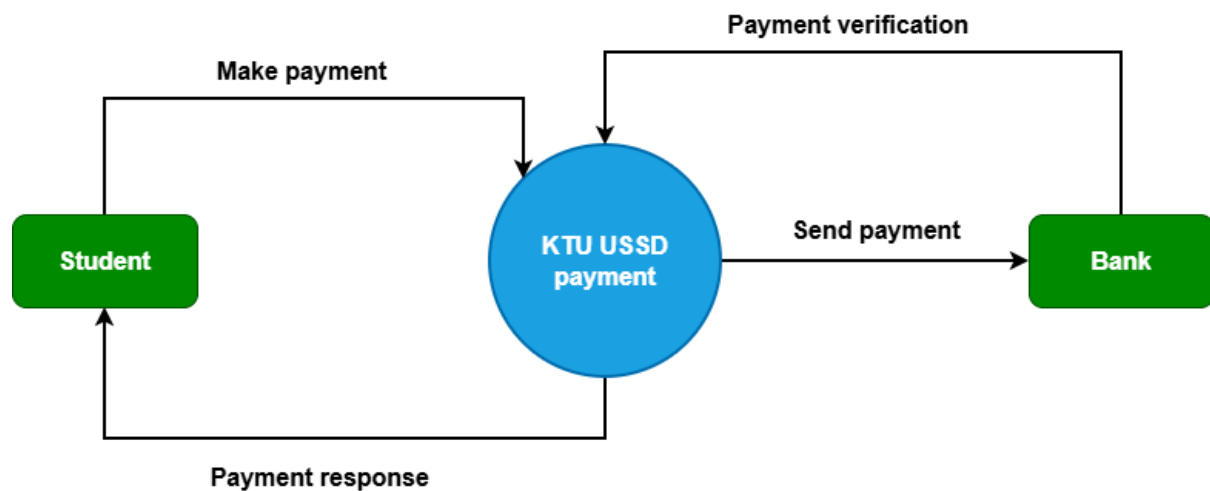
The database stores student information, transaction records, and payment confirmations.

5. Notification Module

This module sends SMS confirmations to students after successful transactions.

3.3.3 Context Diagram of the Existing System

The context diagram illustrates the interaction between the student, the payment system, and external payment services.



3.3.4 Data Flow Description of the Existing System

The system processes payment requests through the following data flow sequence:

1. The student initiates a payment request.
2. The USSD platform validates the student information.
3. The system forwards payment details to the mobile money provider.
4. The provider authorizes the transaction.
5. The bank processes the payment.
6. Transaction records are stored in the database.
7. Confirmation messages are sent to the student and finance office.

3.4 Strengths of the Existing System

The current USSD-based payment system offers several important advantages.

1. Accessibility Without Internet

The system functions without internet connectivity, making it accessible to students using both smartphones and basic mobile phones.

2. Convenience

Students can make payments remotely without physically visiting banks or campus payment offices.

3. Reduced Congestion

The system minimizes long queues at banks and university finance offices during registration periods.

4. Faster Transactions

Payments are processed more quickly compared to traditional manual payment methods.

5. Cost-Effective Operation

USSD transactions generally require minimal operational costs for students and institutions.

6. Broad Device Compatibility

The system operates on all types of mobile devices regardless of operating system or smartphone capability.

3.5 Weaknesses of the Existing System

Despite its advantages, the current system has several limitations that negatively affect usability and efficiency.

1. Limited User Interface

The text-based USSD interface provides poor user experience compared to modern mobile applications. Navigation can become difficult and confusing for users.

2. Session Timeout Issues

USSD sessions expire quickly when users delay responses, causing incomplete transactions and user frustration.

3. Lack of Transaction History

The system does not provide students with a centralized platform for viewing previous payment records and transaction history.

4. Dependence on SMS Confirmations

Proof of payment depends mainly on SMS notifications, which may be accidentally deleted or lost.

5. No Downloadable Receipts

Students cannot generate or download PDF receipts for future reference or official documentation.

6. Limited Scalability

The USSD platform has limited capability for integrating advanced features such as dashboards, analytics, and personalized financial management tools.

7. Security Limitations

Although USSD provides basic security, it lacks advanced encryption and authentication mechanisms commonly found in mobile applications.

8. Poor Error Handling

Users often receive unclear error messages when transactions fail, making troubleshooting difficult.

3.6 Comparative Study of Existing Systems

This section compares the current KTU USSD payment system with Pay4Me and SchoolPay systems reviewed in Chapter Two.

Feature	KTU USSD	Pay4Me	SchoolPay
Internet Requirement	No	Yes	Yes
Smartphone Requirement	No	Yes	Yes
Transaction History	Limited	Available	Available
Downloadable Receipts	No	Yes	Yes
User Interface	Text-based	Graphical Mobile App	Graphical Mobile App
Multiple Payment Options	Limited	Yes	Yes
Real-Time Notifications	Yes	Yes	Yes
Security Features	Basic	Advanced	Advanced
Payment Flexibility	Moderate	High	High
User Experience	Low	High	High
Financial Analytics	No	Limited	Yes

The comparison shows that although the KTU USSD system performs well in accessibility and offline functionality, it lacks many advanced features available in modern mobile payment applications. Systems such as Pay4Me and SchoolPay provide enhanced user experience, permanent transaction history, downloadable receipts, improved security, and better scalability.

However, these systems depend heavily on internet connectivity and smartphone usage, which may limit accessibility for some users. Therefore, the proposed Student Pay Mobile Application aims to combine the accessibility advantages of the USSD system with the advanced functionalities of modern mobile payment platforms.

3.7 Crystallization of the Research Problem

The analysis of the existing payment system at Koforidua Technical University reveals several operational and technological limitations that affect students and university administrators. Although the introduction of the USSD-based payment platform improved accessibility and reduced dependence on traditional bank payments, the system still lacks several important functionalities required in a modern digital financial environment.

Students experience difficulties associated with session timeouts, poor navigation, lack of transaction history, absence of downloadable receipts, and limited payment tracking capabilities. Additionally, the reliance on SMS confirmations creates risks of losing proof of payment. The system also provides minimal interaction and personalization, limiting user engagement and overall satisfaction.

From an administrative perspective, the current system lacks advanced reporting, analytics, and scalable integration capabilities that could improve financial monitoring and operational efficiency. The limited user interface and technological constraints of USSD systems make it difficult to implement more sophisticated financial services and user support features.

The comparative analysis conducted in this chapter demonstrates that modern payment applications such as Pay4Me and SchoolPay provide more robust solutions through graphical interfaces, real-time tracking, downloadable receipts, transaction management, and enhanced security. However, these systems are not specifically tailored to the operational needs of Koforidua Technical University.

Therefore, there is a need for a dedicated Student Pay Mobile Application that will:

1. Provide a user-friendly mobile interface for students.
2. Integrate multiple digital payment options.
3. Generate instant downloadable PDF receipts.
4. Maintain secure transaction histories.

5. Improve transparency and accountability in fee payments.
6. Provide real-time transaction notifications.
7. Enhance overall efficiency in university financial operations.

The proposed system seeks to address these identified gaps by providing a secure, scalable, and user-centered mobile payment solution specifically designed for students of Koforidua Technical University.

3.8 Chapter Summary

This chapter examined the existing fee payment system used at Koforidua Technical University and identified the major operational challenges associated with the current USSD-based approach. The chapter discussed the history, operational procedures, components, strengths, and weaknesses of the system.

A comparative analysis between the KTU USSD system, Pay4Me, and SchoolPay was also conducted to identify gaps in current payment solutions. The findings revealed that while the existing system improves accessibility, it lacks advanced functionalities such as transaction history management, downloadable receipts, enhanced security, and modern user experience features.

These identified limitations justify the need for the proposed Student Pay Mobile Application, which aims to provide a more efficient, secure, and user-friendly digital payment solution for students.